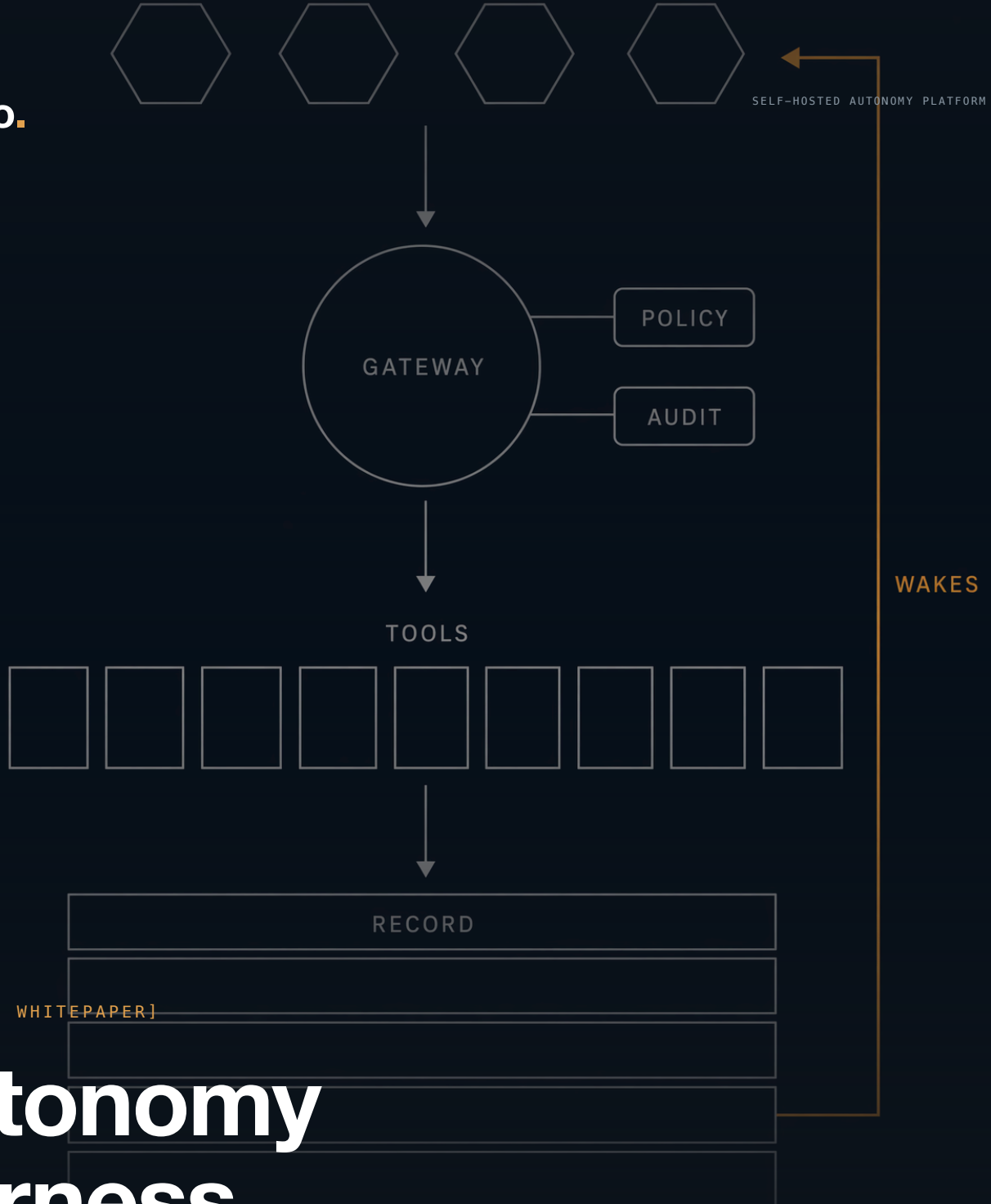




[TECHNICAL WHITEPAPER]

Autonomy Harness

A reactive and proactive cognition stack: autonomous agents that run live operations — on infrastructure you own.

VERSION 1.1 · JULY 2026

[TRANSCRIPT — A HYPOTHETICAL MISSION]

LIVE TRAFFIC WORLD · ECLIPSE SUMO, SIMULATION OF URBAN MOBILITY · LUXEMBOURG SCENARIO

06:41:02 **wake** timer · hourly review
06:41:03 **context** slice: 2 active missions · 214 vehicles tracked · signals nominal
06:41:04 **episode** #4,112 closed — no action required · sleeping

OPERATOR → PILOT

New mission: survey the northern district for congestion buildup between 07:00 and 09:00, and keep it moving. Boundary attached (survey-north.kml); intersection priorities from ops (priorities.xlsx). Interventions stay inside city signal policy. Report with evidence.

06:42:07 **wake** operator message · 2 attachments
06:42:08 **reason** two formats I have never seen from this operator — a .kml boundary, an .xlsx priority sheet; I can structure both
06:42:09 **act** **duckdb_ingest** priorities.xlsx → ops_priorities · 14 intersections ranked
06:42:10 **act** **map_validate_geofence** survey-north.kml → WGS84 polygon · 8.2 km²
06:42:11 **act** **memory_write** missions ← 'survey-north' · objectives, boundary, constraints
06:42:12 **reason** evidence means cameras and counts — four streams cover the district
06:42:13 **act** **subscribe** sumo://congestion · scoped to the boundary
06:42:14 **act** **schedule** wake 07:00 · survey window opens
06:42:15 **episode** #4,113 closed — the mission is now part of the **world model** · sleeping

07:00:00 **wake** timer · survey window open
07:00:03 **act** **query_recording** latest-at → 61 vehicles in boundary · edge N-14 densest
07:00:05 **act** **stream_run_recording** /world/camera/n14 · task t-2211 · sleeping
07:11:40 **wake** task result · 1,204 detections · 87 tracks · northbound queue +9%/min
07:11:43 **act** **duckdb_ingest** detections → demand per approach
07:11:44 **act** **optimization_solve_milp** · minimize delay · typed signal choices, demand, priorities, and policy constraints
07:11:52 **reason** verified solution retimes N-14 and N-9 inside policy; leaves N-2 alone — low priority, low demand; the numbers say so
07:11:53 **act** **sumo_set_signal_phase** N-14 N-9 · run_batch(500) · task t-2212 · sleeping
07:19:26 **wake** task result · queue at N-14 down **23%** · writing report
07:19:31 **report** artifact sealed — counts, plan, clip, rationale · shareable under policy

An illustrative transcript. Every surface in it — the wakes, the tools, the durable tasks, the artifacts — is the shipped system. The rest of this paper is the machinery that makes this ordinary.

Abstract

What an operation needs from autonomy is harder to supply than intelligence. Decisions have to follow from what was actually observed, and they have to stay aligned with the mission — with its objectives and constraints, and with the intent behind them when circumstances force a change of plan. Permission has to hold even when the model's judgment fails; especially then. Resilience has to be engineered: processes die, machines are lost, tasks are cut off mid-flight, and the system must come back knowing what it was doing, with the consequences of every interruption defined in advance. What the autonomy is doing has to be observable while it is doing it. And the operator, who answers for all of it, needs assurance that behavior stays within what was approved — and an account of every action, attributable and replayable from the record.

The **Autonomy Harness** is a self-hosted cognition stack built to supply these guarantees structurally. Its foundation is a **world model**: everything the operation senses, remembers, and decides, fused into a durable body of state that agents query rather than accumulate. The mission is part of that model — objectives, constraints, and current beliefs held as data — so intent is something an agent reasons over, and a deviation is measured against the goal it serves before it is adopted and recorded. Reasoning runs in bounded episodes over fresh slices of this model, which is why an agent's knowledge outlives any single process or context window. Between every agent and every tool sits one gateway, enforcing the operator's policy outside the model's reasoning; a refusal there leaves the same evidence an approval does. Work too long for an episode becomes a durable task whose recovery behavior is declared before it starts. Beneath all of this, a producer-local forwarder holds each signal until Recording Hub acknowledges fsynced journal state and an immutable ordered part. One verified archive pass reduced a representative 30,307-chunk corpus to 31 indexed chunks in 2.56 seconds. The record that grounds a decision can also replay it. New hosted servers attach through a full MCP contract, in Rust or Python, and inherit the task, artifact, identity, policy, and audit guarantees on arrival. A live traffic world exercises the complete loop, and the identical stack runs on a single edge host or inside an air-gapped cluster.

KEYWORDS autonomous agents · world model · runtime governance · durable execution ·
acknowledged recording · edge and air-gap deployment · MCP

[READING GUIDE]

Section 1 states the contributions; the abstract carries the requirements. Sections 2 through 8 develop the architecture. Section 9 validates the loop on a live traffic world; Section 10 positions the platform as self-improvement primitives; Section 11 covers deployment. Appendix A defines every component.



Contents

#	Section
1	Contributions
2	System Model
3	The World Model
4	The Agent Kernel
5	Governance: Edge, Access, and the Gateway
6	The Capability Plane
7	Durable Work and Shared Planes
8	The Record
9	Empirical Validation
10	Self-Improvement Primitives
11	Deployment Considerations
12	Conclusion
A	Appendix: Component Reference



1. Contributions

The Autonomy Harness makes six architectural contributions. Each is developed in the section noted and exercised end to end by the validation in Section 9.

- 1 **The world model as agent context (§3).** Everything the operation senses, remembers, and decides fuses into one durable, queryable model of reality — and the mission lives in it as data, so intent is something an agent reasons over rather than a prompt it was once shown. Episodes draw fresh slices by query; knowledge survives restarts and outlives context windows.
- 2 **Cognition governed from outside the model (§5).** Every action clears one authenticated gateway and answers to installation policy — refused by default when in doubt. A model can be wrong, drift, or swallow hostile instructions and still cannot act beyond the envelope; even refusals leave evidence.
- 3 **Acknowledged recording (§8).** A producer-local forwarder retains a bounded queue until Recording Hub fsyncs journal state, materializes an immutable ordered RRD part, and advances the checkpoint. One IDR-aligned archive pass writes indexed immutable shards.
- 4 **Durable tasks with declared recovery semantics (§7).** Long work detaches from the episode that started it and survives process death. Every operation declares its recovery class — resumable, webhook-bound, or fail-closed — and completion wakes the sleeping agent within a second.
- 5 **A modular capability plane under one contract (§6).** Sixteen hosted server identities ship in the canonical control plane. Any local, bridged, or remote MCP server registers in the validated catalog and immediately answers to the same task, artifact, identity, policy, and audit contracts; domain administration follows the same governed path.
- 6 **Deployment-invariant cognition (§11).** The same platform, the same agents, and the same operational intelligence run on one edge host, across a cluster, sealed inside an air gap, or hybrid — with provenance included and connectivity on the operator's terms.



2. System Model

The system closes one loop. Live encoded media enters Stream directly while an independent forwarder records governed evidence; agents combine live results, authorized replay, and grounded Reason output, then act through governed tools. Effects land back in the world, and every step deposits evidence. The shape is Boyd's OODA loop — observe, orient, decide, act — with orientation held in durable state rather than inside the agent. Data moves along this spine in three ways. Producers push inward while Stream processes arriving frames directly, and agents query authorized recording snapshots back out — observation. Tool calls cross the gateway, and the long ones become durable tasks whose completion wakes a sleeping agent. Whatever remains is the account: authentication decisions, task transitions, artifact grants, and every agent step, landing in the platform store and the decision log.

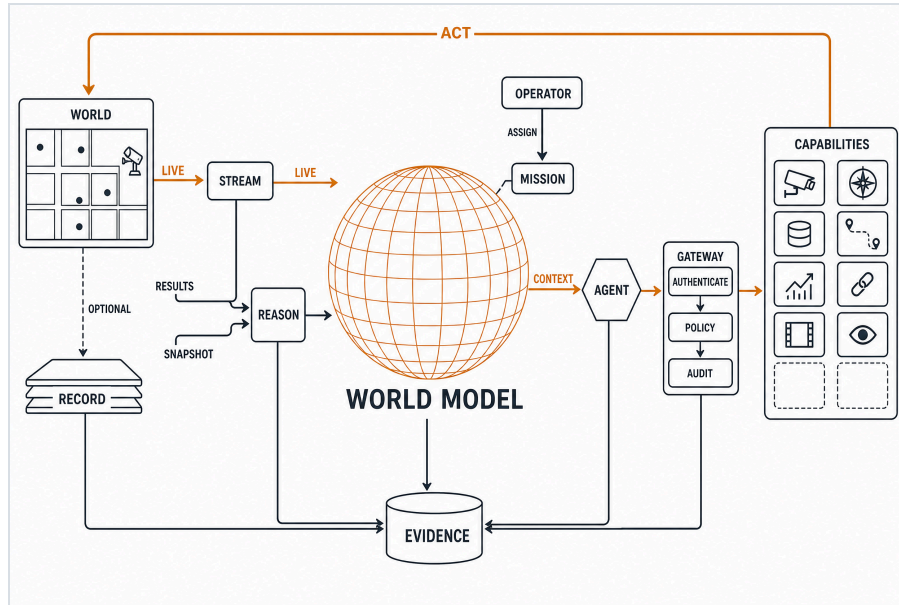


FIGURE 1 – THE AUTONOMY HARNESS: DIRECT LIVE STREAM, INDEPENDENT GOVERNED RECORDING, GROUNDED REASON OUTPUT, AND AGENT ACTION THROUGH ONE IDENTITY, POLICY, AND AUDIT BOUNDARY.

Flow	Path	Guarantee
Observe	Arriving H.264 enters Stream directly; a producer-local forwarder uploads recording batches; agents read immutable task-start snapshots through Recording and Reason.	Live results never wait for Recording Hub; acknowledgement follows fsynced journal state and part materialization.
Act	Every tool call enters through the gateway under one flat namespace; quick tools answer inline; long tools return a durable task handle immediately.	Task completion pushes a wake; the next episode assembles with the result.
Account	Auth, policy, admin, and task transitions write audit evidence; every decision records what the agent saw, chose, and why.	Usage records meter provider spend and tool activity per principal.

3. The World Model

Agent context is a **world model**: everything the operation senses, remembers, and decides, fused into one durable, queryable model of reality. Sensors and simulators stream the physical world into the record; agents keep their beliefs, missions, and constraints in analytical memory; every decision lands in the log — and each act an agent takes becomes part of the model the next episode reasons from. Context is assembled fresh from the world model every episode, so an agent's knowledge survives any restart and outlives any context window. In OODA terms, the world model is the Orient stage made durable and queryable.

The mission is part of the model, held as data rather than prompt text. Objectives, constraints, and current beliefs occupy tables the agent reads and writes through guarded instruments, which makes intent something it can reason about: a proposed deviation gets measured against the objective it serves, adopted deliberately, and recorded with its rationale. Adaptation stays subordinate to the mission because both live in the same queryable state.

A mission does not have to arrive in any particular format. Whatever the source produces — a scenario file from another autonomy stack, an exported plan, an annotated map, a plain document — the agent reads it, interrogates it, and structures it into the model: objectives and constraints become rows in its tables, geometry passes through the Map and Frames servers into real coordinates, and the original lands in the artifact plane with its provenance intact. Integration with an existing stack begins with handing the agent its files.

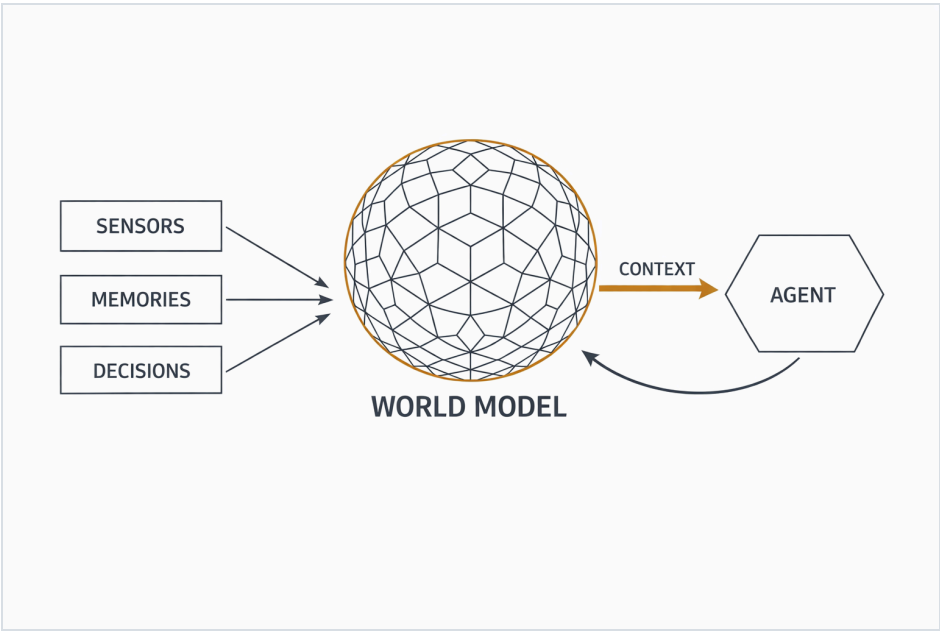


FIGURE 2 — SENSORS, MEMORIES, AND DECISIONS FUSE INTO THE WORLD MODEL; EVERY EPISODE'S CONTEXT IS A FRESH SLICE OF IT.

Source	What it contributes	What agents draw
Sensors & sims	The physical world, streamed into the durable time-and-space record.	The world now (latest-at) · how it got here (trajectories and ranges).
Agent memory	Beliefs, missions, constraints, and state in each agent's analytical memory.	What the agent believes and intends — queryable with guarded SQL.
Decision log	Every turn, tool call, and rationale, time-indexed and replayable.	What happened and why — the model of the agents themselves.

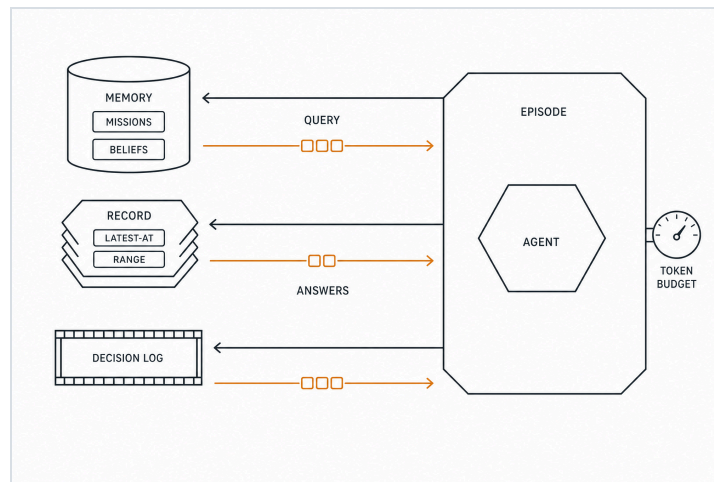


3.1 Context, Queried

The world model changes what an episode's context is made of. Context here is assembled by query: the kernel composes each episode from manifest-defined slices, and the agent holds the same instruments, so mid-episode it asks for exactly what it needs — rows from its own memory, a snapshot or trajectory from the record, analytics over anything it has ingested.

Instrument	Example	What the episode gets
memory_query	SELECT id, eta, risk FROM missions WHERE status = 'active'	Exactly the active missions, as rows; beliefs and constraints join the same way.
query_recording · latest-at	every vehicle's position, now	A world snapshot, measured — the state of reality in one call.
query_recording · range	vehicle 214, last five minutes	One trajectory, on the original timeline.
timeline_query	my tool calls and outcomes, last six hours	The agent reads its own conduct as data.
duckdb_query	SELECT class, count(*) FROM detections GROUP BY window_5m	Analytics over anything ingested — including perception output.

The strategy carries for two reasons. Tokens spend on answers rather than history, so a mission's entire history compresses to the rows this episode needs. And because every instrument reads the same governed planes, results join — detections land in SQL, SQL rows feed planning, committed plans write back to memory. Section 6.1 shows the composition end to end.



THE EPISODE QUERIES THE PLANES — MEMORY, RECORD, DECISION LOG — AND ONLY ANSWERS COME BACK TO SPEND TOKENS, UNDER A HARD BUDGET.

4. The Agent Kernel

An agent runs indefinitely because durable stores are the memory: every episode's context is assembled fresh from the world model, and everything in flight survives process death. The kernel validates every manifest against its schema before execution. An agent type combines a manifest, a preamble, and SQL migrations. Fielding another agent is an act of configuration, and one installation runs one agent or a team of them.

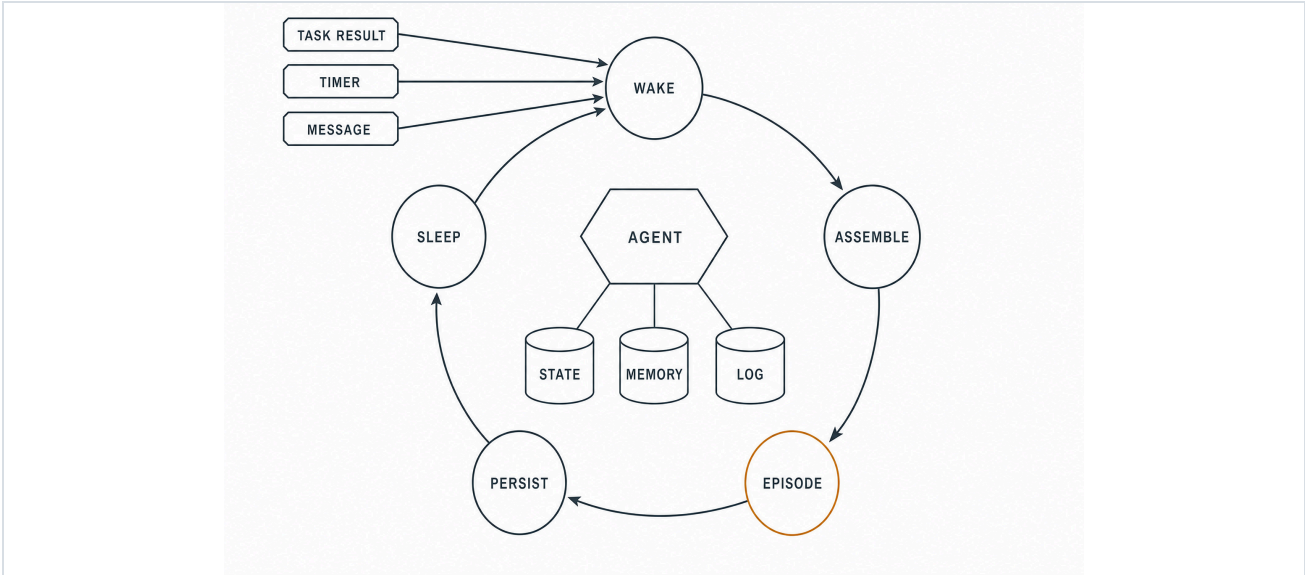


FIGURE 3 – THE LIFECYCLE: WAKE → ASSEMBLE → EPISODE → PERSIST → SLEEP, GROUNDED ON THE DURABLE PLANES.

Stage	What happens
Wake	A batch of wakes, coalesced and deduped; every wake is persisted with its disposition, so the ledger explains why each episode ran.
Assemble	A fresh slice of the world model — current state, beliefs, and history — plus the wake payload, under a token budget.
Episode	One bounded reasoning run. Quick gateway tools answer inline; long tools dispatch as durable tasks. Hooks record every step and terminate over-budget runs.
Persist	Unresolved task descriptors land in the platform store; the episode closes with usage counters and a summary.
Sleep	Idle on the wake bus. A leased watcher per detached task waits on the result; the heartbeat bounds silence.

Wakes: task result · resource change · timer · operator message · input response. A wake survives process death — descriptors and watcher leases rehydrate from the platform store on any boot, so kill and restart resumes the same tasks.



4.1 Memory Planes and Durability

Memory planes

- **Scheduling truth.** The platform store owns episodes, wakes, task watchers, and leases; the transactional outbox and changefeed deliver wakes across processes.
- **Analytical memory.** A sandboxed analytical database per agent: domain tables (the Pilot: targets, missions, waypoints, constraints, beliefs), a kv store, and an episode-log projection for context assembly.
- **Decision log.** An append-only recording: every turn, tool call, task lifecycle, prompt, and summary — time-indexed and watchable live.
- The agent reads all three itself: `memory_query` (guarded SQL), `memory_write` (bounded mutations), and `timeline_query` (dataframe reads over the log).

Durability, measured

- Detach at episode end: a task's descriptor outlives the process; any later boot rehydrates the live handle and keeps waiting.
- Token rotation is make-before-break; task ids are principal-scoped, so continuity holds across reconnects.
- The wake is push, arithmetically: with polling pinned at 60 s and the heartbeat at 600 s, a validation agent slept a task's full ~15 s and woke within one second of completion — verified live with a real model end to end.
- Budget hooks book overruns as `budget_terminated`, a first-class outcome; hourly windows defer non-priority wakes.

[PRINCIPLE]

Agents are data. At boot, schema validation rejects an invalid manifest. The manifest declares the model endpoint, gateway principal, episode limits, budgets, schedule, context sections, and writable tables. A preamble and domain migrations complete the profile; cross-episode instructions live there or in memory.



5. Governance: Edge, Access, and the Gateway

Every client enters through the installation's own ingress, and every action runs the same gauntlet: authenticate, then policy, then audit. Enforcement lives outside the agent's own reasoning — a model can be wrong, drift, or swallow hostile instructions and still cannot act beyond the envelope the operator set. The only path onward carries a gateway-signed identity, and every hosted server lives on the internal network behind it.

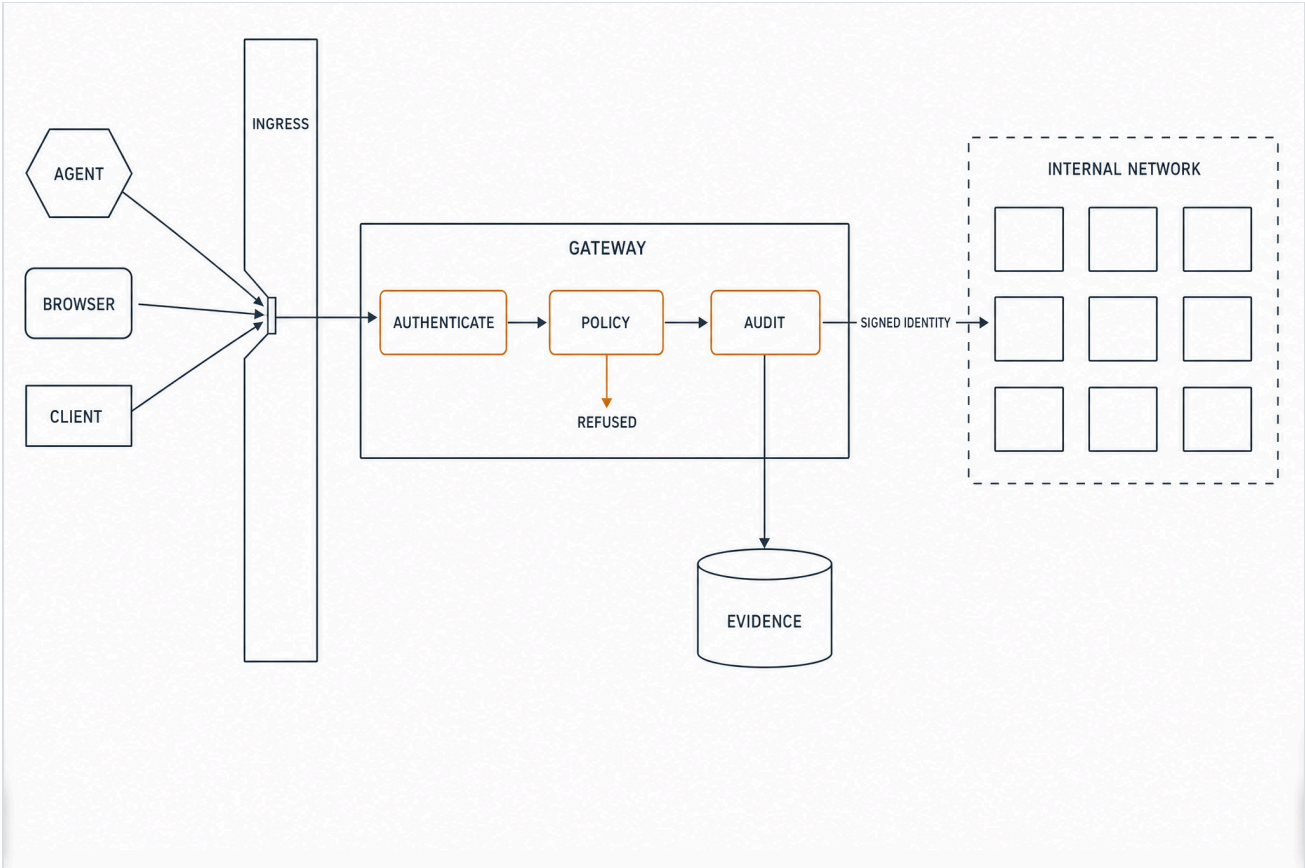


FIGURE 4 – ONE DOOR IN, AUTHENTICATE → POLICY → AUDIT, REFUSED BY DEFAULT, SIGNED IDENTITY ONWARD, EVIDENCE KEPT.

Function	Behavior
Authentication	Browser OAuth with PKCE, client credentials with private-key JWT, identity-assertion grant exchange, JWKS, token issuance and revocation, durable refresh rotation — reuse revokes the family.
Policy & audit	Attribute- and role-based policy over tools, resources, tasks, and artifacts; tenant and data-label checks for regulated data; fail-closed decisions with reason codes; a durable audit trail.
Control plane	Validated server / profile / policy catalog, authenticated live apply and reload, secret references with redaction, an admin API for the console, and policy-checked, audited artifact downloads.
Domain administration	Scoped MCP tools, resources, durable tasks, and Apps are canonical. A declared /admin/{profile}/servers/{server}/{path} route is an additive projection over the same typed models, policy, audit, identities, and state. Map administration is MCP-native.
Protocol surface	Tools, resources and templates, prompts, completions, tasks, subscriptions, notifications, structured content with declared schemas, artifacts, and usage. Tool names gain a prefix only at aggregation; resource URIs retain their owning scheme.



6. The Capability Plane

The canonical control plane defines sixteen hosted server identities. Each server declares the MCP surfaces its domain needs and runs behind gateway-signed identity; a new local, bridged, or remote endpoint registers in the validated catalog and immediately answers to the same task, artifact, identity, policy, and audit contracts. The gateway discovers the declared surface, prefixes tool names only at aggregation, and preserves each resource URI scheme. A client connects once and reaches every capability; a system joins once and reaches every client.

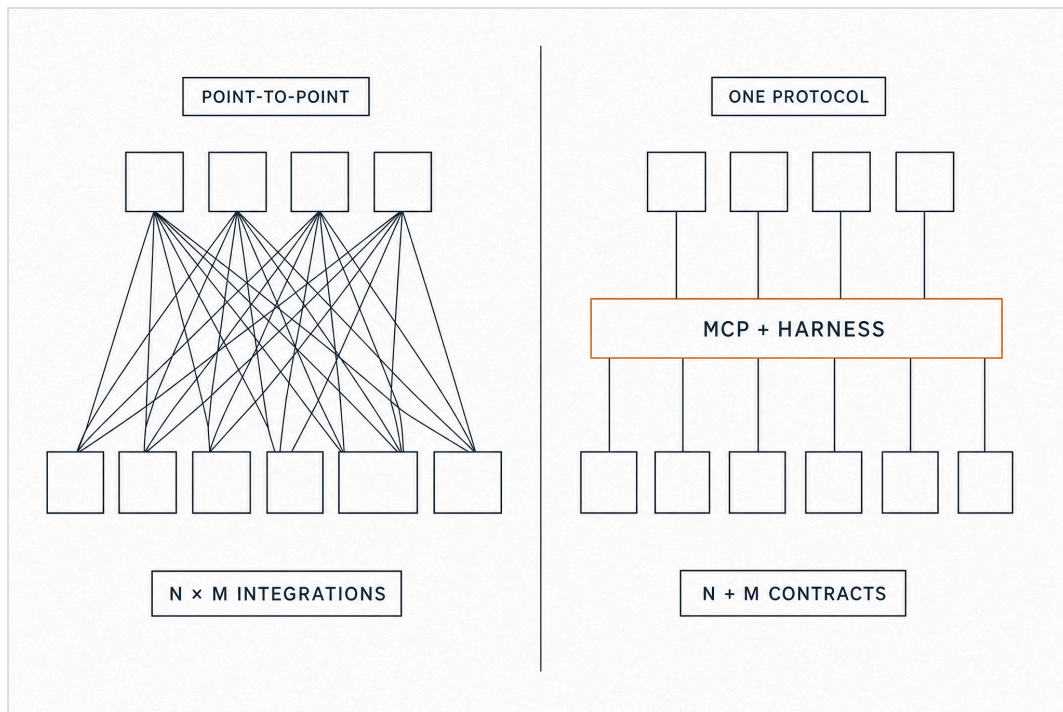


FIGURE 5 — FROM $N \times M$ INTEGRATIONS TO $N + M$ CONTRACTS.

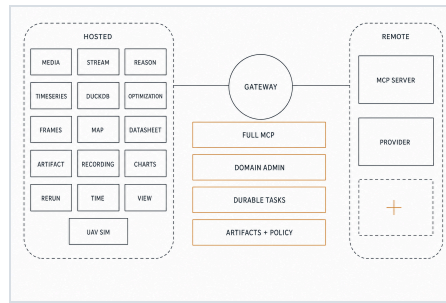


FIGURE 6 – SIXTEEN HOSTED SERVERS AND REMOTE ENDPOINTS; FULL MCP, CANONICAL DOMAIN ADMINISTRATION, DURABLE TASKS, ARTIFACTS, AND POLICY REMAIN ONE GOVERNED SYSTEM.

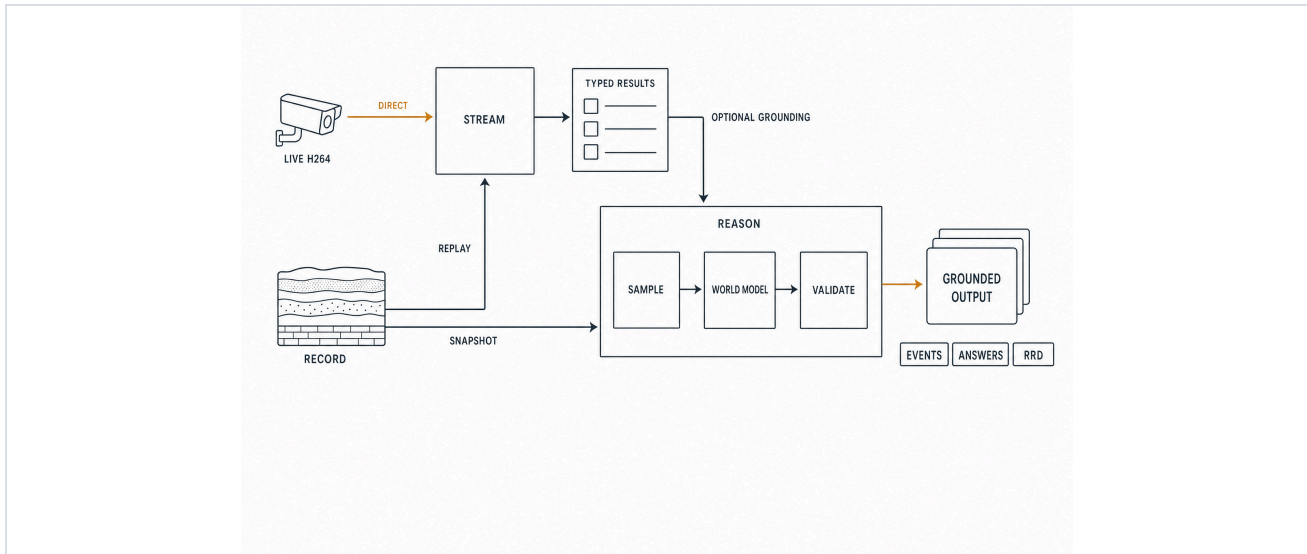
Server	Capability	Representative tools
Media media	Provider-backed generative media with webhook-driven completion, a model catalog, and pricing.	<code>run · models · model_schema · artifact</code>
Stream stream	Direct live RTP/H.264 processing plus reproducible replay over authorized task-start snapshots, with encoded App preview and typed results.	<code>start_live_session · stop_live_session · run_recording</code>
Reason reason	Local world-model reasoning over recording snapshots, optionally grounded by governed Stream perception results.	<code>analyze_recording</code>
Forecasting timeseries	Forecasting over any SQL-readable time-series source, visualized as recordings.	<code>forecast</code>
Time time	Authority-bound civil, military, GNSS, and mission time; calendars, timelines, clock quality, and temporal events.	<code>resolve_time · convert_time · assess_clock · evaluate_windows</code>
SQL duckdb	Arbitrary SQL over principal-owned databases inside a locked, resource-bounded sandbox.	<code>query · execute · ingest · export</code>
Optimization optimization	NVIDIA cuOpt vehicle routing, scenario batches, convex and MILP solving, with immutable problem, run, solution, and verification evidence.	<code>optimize_routes · optimize_route_scenarios · solve_convex · solve_milp · verify_solution</code>
Frames frames	Complete ECEF-rooted frame worlds, immutable revisions, bounded conversion, durable batches, and provenance.	<code>create_world · publish_world · convert_frame · batch_transform</code>
Map map	Earth geography, governed releases and restrictions, Valhalla and governed-network routes, cuOpt travel models, and reachable areas with pinned provenance.	<code>search_locations · route · route_matrix · build_travel_model · reachable_area · publish_restriction · validate_route</code>
Datasheet datasheet	Python reference server for dataset preview, column statistics, and durable artifact-backed profiling.	<code>preview_dataset · column_stats · profile_dataset</code>
Sharing artifact	Discovery, metadata, grants, release, and sharing — the governance surface over the byte store.	<code>metadata · grant_access · revoke_access · set_release_state · create_share_link · revoke_share_link</code>
Recordings recording	Authorized access to the record: discovery, dataframe queries, subscriptions, publication.	<code>query_recording · seal_recording</code>
Charts charts	Chart rendering projected through the gateway, turning query results into shareable visuals.	<code>render_chart · compile_chart · validate_chart · list_chart_types · create_chart_view</code>
Viewer rerun	A live viewer driven entirely over MCP — session, capture, inspection, and UI automation.	<code>17 interactive tools (connect ... batch)</code>
3D View view	Governed static scene compositions with exact inputs and bounded overlays, rendered offscreen on NVIDIA Vulkan.	<code>create_scene_composition · create_view · set_camera · capture_frame · close_view</code>
UAV Simulation uav-sim	Governed sessions, missions, vehicles, terrain, dataset capture, authoritative logical cameras, isolated native GPU products, per-viewer WebRTC leases, and provider-neutral simulator control.	<code>configure_world · execute_mission · list_live_cameras · open_live_view</code>

6.1 In Depth: Live Stream, Replay, and Grounded Reason

Stream begins processing arriving RTP/H.264 frames immediately. Its encoded tee feeds a bounded App preview and may feed the independent recording forwarder, but live results never wait for Recording Hub. The same admitted pass-through or perception profile can run later over one immutable task-start snapshot of complete acknowledged recording parts.

- **Live:** start one owner-scoped session, process new frames on the GPU, and publish typed detections while the H.264 App draws overlays.
- **Replay:** run the same approved profile over an authorized recording range and publish immutable JSON and RRD results.
- **Reason:** combine an authorized recording snapshot with a bounded typed subset from `stream://{id}`; answers may cite Stream track identities.

Stream processing, Reason inference, and visual acceptance fail closed without their declared hardware. Browser H.264 software decode is allowed only when Media Capabilities reports the exact configuration as supported and smooth, and the App labels that path as software.



LIVE STREAM AND AUTHORIZED REPLAY PRODUCE GOVERNED RESULTS; REASON COMBINES AN IMMUTABLE RECORDING SNAPSHOT WITH OPTIONAL TYPED STREAM GROUNDING.



6.2 In Depth: Optimization

Optimization handles vehicle routing and bounded mathematical decisions. Routing assigns orders to heterogeneous vehicles and sequences every stop. Convex and mixed-integer models express decisions that do not fit a route contract. The kinds of questions it answers:

- Three survey drones, forty waypoints, battery limits — which drone flies which route?
- Given measured demand per approach, which signal-timing plan minimizes total delay?
- Ten targets, two sensors, shifting priorities — what gets watched this hour?

Map publishes immutable cost and transit-time matrices for each vehicle type. Agents may use SQL observations to formulate coefficients, but the submitted problem is controlled JSON with stable location, order, vehicle, variable, and constraint identities. The result is an immutable solution linked to the exact problem and execution run.

The Rust control plane validates and compiles the typed model before NVIDIA cuOpt 26.06 runs it on a required GPU. Routing supports service or pickup-delivery orders, windows, breaks, capacities, optional orders, restrictions, warm starts, and weighted objectives. Mathematical work covers continuous LP, QP, QCQP, and SOCP representations plus linear MILP. A separate verifier recalculates route feasibility, bounds, integrality, constraints, and objective values before the solution becomes evidence.

One governed loop

#	Call	Result
1	wake — resources/updated: sumo://congestion	A jam forming is a wake, the moment it happens.
2	query_recording · latest-at	World snapshot: edge E14 saturated.
3	stream_run_recording /world/camera/e14 · 10 min	1,204 detections · 87 tracks — cars, buses, trucks, counted per approach.
4	duckdb_ingest → query	Demand per approach, as SQL rows.
5	optimization_solve_milp · typed signal choices and policy constraints	A GPU-solved signal-timing solution with independent feasibility and objective verification.
6	sumo_set_signal_phase · run_batch(500)	Applied; the agent sleeps through the batch.
7	wake — task result	Delay –23%; every step already in the record.



7. Durable Work and Shared Planes

Durable tasks

Every long-running tool runs on the shared task runtime. Task identities are time-ordered; creation, leases, transitions, results, and outbox events are atomic in the platform store. Recovery is declared per operation, so an interruption always has defined, recorded semantics:

Class	Meaning
<code>resume</code>	Deterministic, side-effect-safe work may be reclaimed after its lease expires.
<code>webhook_wait</code>	A durably submitted provider job waits for its signed webhook. Completion is webhook-only.
<code>interrupted_indeterminate</code>	Interrupted mutating work fails closed and is never replayed automatically.

Shared planes

- **Platform store.** One durable coordination authority for identity, policy, control-plane revisions, tasks, artifacts, recordings, map catalogs and releases, agents, wakes, audit evidence, and usage. The transactional outbox and replayable changefeed carry all cross-process work.
- **Artifact plane.** A single byte-level policy-enforcement point: object storage for bytes, the platform store for grants, labels, and release state. Fresh opaque identity per occurrence; hashes provide integrity and tenant-local dedup; artifact references let one server consume another's output.
- **Contract core.** The provider-neutral contract every server shares: Rust newtypes and enums, deployment models, exported JSON Schemas, and gateway-signed identity assertions — all validated fail-closed; servers hold public verification keys only.

[PRINCIPLE]

Two sharing modes, both explicit: authorized grants to users or groups, still constrained by tenant and label policy; and read-only anyone-with-link bearers for artifacts explicitly marked releasable — expiring, download-limited, and revocable.



8. The Record

Each producer publishes native Rerun traffic to a loopback forwarder with a bounded persistent queue. The forwarder authenticates through the gateway and resumes versioned protobuf upload from the Hub checkpoint. Recording Hub fsyncs journal state and materializes an immutable ordered RRD part before acknowledgement. One cataloged writing segment closes at one hour or the 192 MiB precompaction safety limit, aligned to a decoder-reentrant IDR. Archive publication is one verified footer-writing object-store pass. The record remains the agents' ground truth and the operation's evidence chain, while direct live Stream processing stays independent from Hub latency.

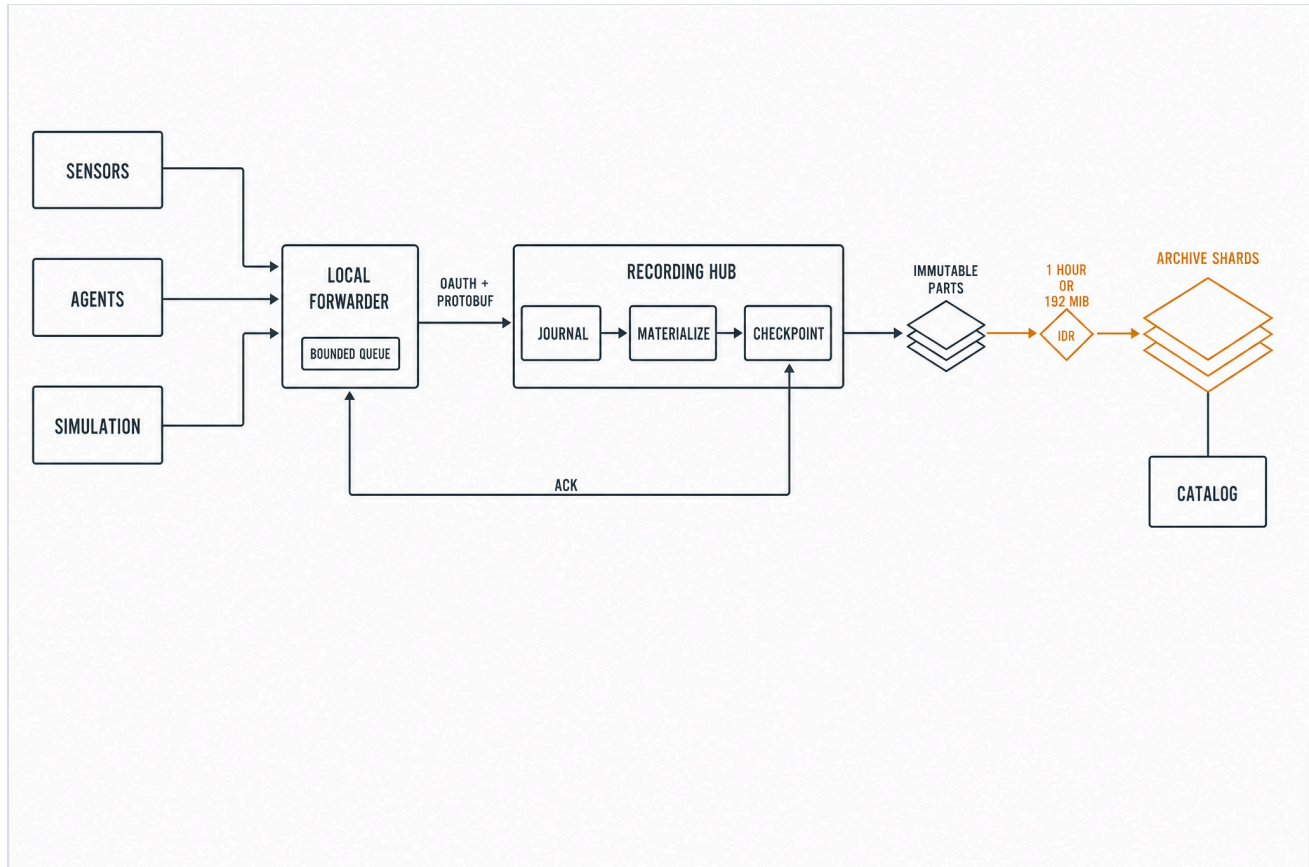


FIGURE 7 – DURABLE CAPTURE: FSYNCD JOURNAL → CHECKPOINT → IMMUTABLE PARTS → INDEXED SHARDS. A REPRESENTATIVE 30,307-CHUNK CORPUS BECAME 31 ARCHIVE CHUNKS IN 2.56 SECONDS.



9. Empirical Validation

The claims above are exercised on a live traffic world: SUMO — Simulation of Urban Mobility, an open-source traffic microsimulator — runs the pinned, MIT-licensed LuST Luxembourg scenario in its own container; one sumo-mcp server owns the single serialized control connection, pushes the world into the record as Rerun streams, and exposes traffic control as governed tools. The long operations are durable tasks the agent detaches from and wakes on — the same sleep/wake the kernel runs everywhere, now driven by a real simulator.

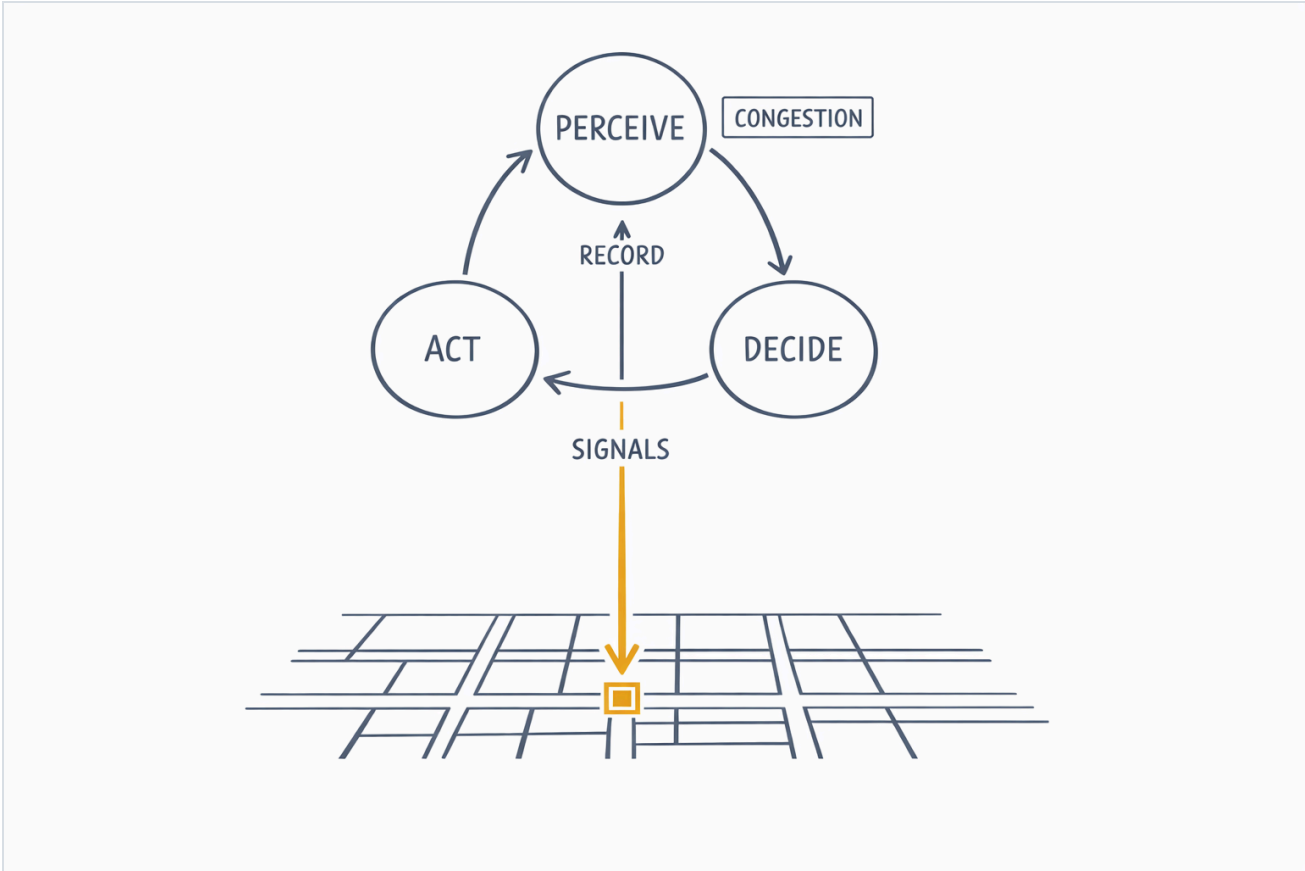


FIGURE 8 — THE CLOSED LOOP ON A LIVE CITY: PERCEIVE FROM THE RECORD, DECIDE, ACT ON SIGNALS AND ROUTES.

Surface	Detail
Sync tools	query_state · describe_scenario · set_signal_phase · reroute_vehicle · set_edge_speed · close_lane · open_lane
Durable tasks	run_batch (live sim — fails closed on interruption) · generate_network · compute_routes · optimize_signals (real SUMO programs, resumable)
Events	sumo://congestion is subscribable — a jam forming is a wake the moment it happens.
Outputs	Successful task outputs upload through a task-bound write capability to the shared artifact plane.



9.1 The Loop, Measured

The exact detach → sleep → wake path: the agent calls a task-augmented tool, receives a durable handle immediately, persists the descriptor, ends the episode, and sleeps. The server runs on; completion pushes a wake, and a fresh episode assembles with the result.

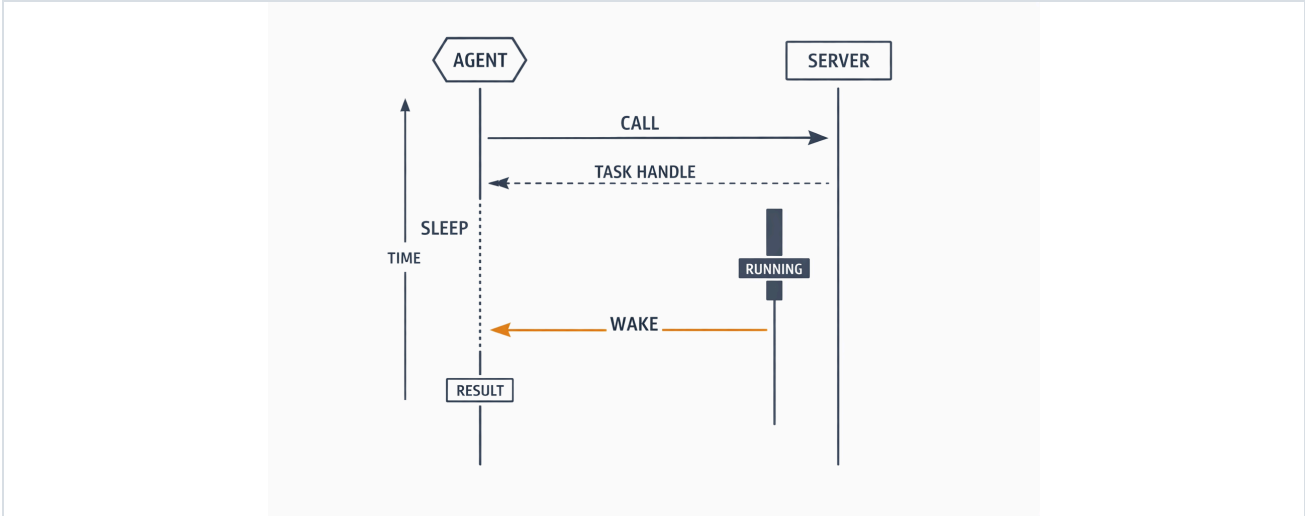


FIGURE 9 – DETACH → SLEEP → WAKE: ZERO CONNECTIONS HELD, ZERO EPISODES BURNED WAITING.

Property	Measured behavior
Archive materialization	76,094,593 source bytes and 30,307 one-row chunks became 31 archive chunks and 28,815,516 bytes in 2.56 seconds, using 138,812 KiB peak RSS.
Video indexing	194 H.264 GoPs and 3,875 frames became 12 IDR-aligned video chunks, each no larger than 2 MiB.
Wake latency	With polling pinned at 60 s and the heartbeat at 600 s, the agent slept a ~15 s task in full and woke within one second of completion — push, arithmetically.
Process death	Task descriptors and watcher leases rehydrate on any boot; kill and restart resumes the same tasks with memory intact.

Methodology

- **Smoke harness.** End-to-end scenarios with real process orchestration: fake identity provider, provider, and upstream services; a real enterprise IdP realm; live agent sleep/wake; and the full traffic world in a disposable project. The Map scenario acquires and activates governed fixtures, runs real Valhalla and network routes through the Task API, and verifies that restrictions and release changes invalidate dependent resources.
- **Conformance.** A generic MCP client exercises the full protocol surface, plus auth-metadata discovery and token tooling; continuous integration runs Rust and Python suites, the datasheet reference server, the console build, the IdP integration path, and deployment contract checks on every change.



10. Self-Improvement Primitives

An operation running on the harness produces, as a side effect, exactly what learning consumes. Every episode leaves a complete trajectory — the context the agent saw, the action it took, and what happened next — time-aligned with the world's own record. The harness does not claim the learning program: it does not train models, tune policies, or decide what counts as better. What it supplies is the substrate a program of reinforcement learning, imitation, or scenario-driven improvement is built from, present from the first deployment.

Primitive	What the platform guarantees	What can be built on it
Complete experience	Every episode records its context slice, tool calls, outcomes, and rationale, aligned to the same timeline as the world's sensor record.	Trajectory datasets — state, action, outcome — mined by query; reward signals computed after the fact from measured results.
Derived ground truth	Stream perception turns live or replayed video into typed facts and RRD annotations on the source timeline.	Training data extracted from operations, with provenance; labels that survive audit.
A world to practice in	The same contracts drive live and simulated worlds; recorded anomalies replay as scenarios.	Policy evaluation and learning rollouts against replayed or simulated reality, off the live system.
Measurable objectives	Missions, constraints, and outcomes live as queryable state; usage and results are metered per principal.	Objective and reward functions defined over data the operation already keeps.
Governed rollout	Agents are data — a candidate variant is a manifest — and every variant acts through the same gateway, budgets, and policy as anything else.	Shadowed or staged variants whose failures are contained by construction, promoted as governed change.
Lineage	Artifacts carry provenance; control-plane revisions and the audit trail record what changed, when, and on whose authority.	A defensible chain from data to training run to deployed variant.

What remains the learning program's to design — the algorithm, the training infrastructure, the criteria for promotion — designs against a platform where experience is already durable, objectives are already measurable, and rollout is already governed.

[POSITION]

The platform does not train models; it makes experience durable, objectives measurable, and rollout governable. Learning is designed on top — and the primitives it needs are already in place.



11. Deployment Considerations

Deployment is a spectrum, and cognition is constant across it. The same platform runs at the operation's edge on one host, across a cluster, sealed inside an air gap, or hybrid — an edge installation that reaches remote capability when links allow. Wherever it lands, it carries the full stack: one agent or a team of them, every hosted capability server, and the operational intelligence to run the mission — the console, the record, and the audit trail.

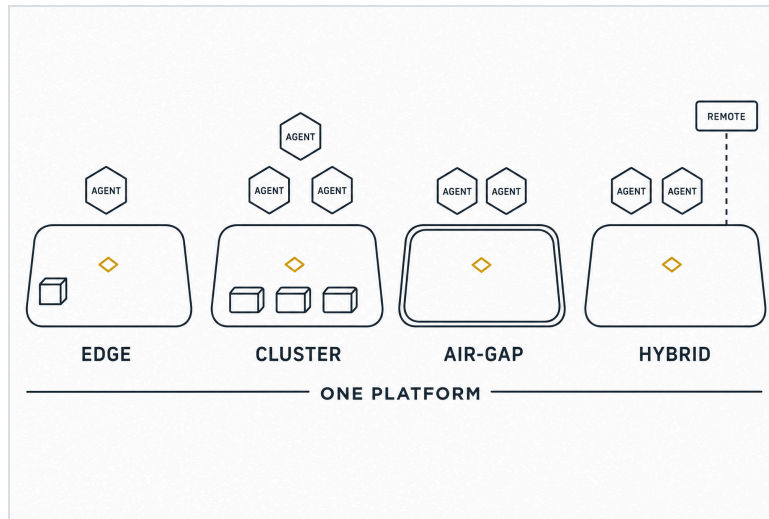


FIGURE 10 — SAME PLATFORM, SAME COGNITION — EDGE, CLUSTER, AIR-GAP, OR HYBRID; ONLY HYBRID REACHES OUT, AND ON THE OPERATOR'S TERMS.

Form Shape

Edge One host at the operation: a single command brings up the full stack behind a loopback edge. Public exposure is the operator's ingress decision.

Cluster The Helm form: one platform-store StatefulSet, default-deny network policy, optional service-mesh mTLS, separate migration and runtime credentials, operator-supplied telemetry and SIEM wiring.

Air-gap The verified offline bundle: pinned images, installation material, configuration schemas, checksums, and SBOMs. Installation and verification complete entirely inside the boundary.

Hybrid An edge installation that registers remote MCP endpoints and provider-backed servers. The reach is governed like everything else — same policy, same audit, same task contract.

12. Conclusion

The Autonomy Harness closes the loop that persistent autonomy requires. An agent here reasons from a world model that carries the operation's history and its mission alike. The gateway holds the operator's policy whatever the model believes. Work survives process death because its recovery was declared before it started, and evidence accumulates as a side effect of running rather than as a reporting obligation afterward. A live traffic world exercises all of it, and the platform deploys unchanged from a single edge host to an air-gapped cluster.

The same contracts are the foundation for what comes next. Teams of agents can coordinate over a shared world model; richer hosted servers can register from the ecosystem; guarantees can be expressed over mission outcomes rather than individual actions. Each grows on the spine this paper described.

[THESIS, RESTATED]

Ground the agent in a world model that carries the operation and its mission. Enforce the rules outside the model. Declare how work fails before it starts, acknowledge evidence only after it is durable, and keep live processing independent. One operator owns all of it, anywhere the mission runs.



Appendix A: Component Reference

Component	Role	What it is
ingress	ours	The single public door, owned by the operator. One place to defend and observe.
gateway	ours	The governed protocol boundary: authenticates the caller, checks policy, projects full MCP server surfaces, carries declared additive administration projections, and writes the audit trail.
console	ours	The cockpit: the operations UI plus a session backend that owns browser login, encrypted sessions, and CSRF enforcement.
agent kernel	ours	The runtime for long-lived agents: short episodes, durable memory, wake on a result, a timer, or a message.
hosted MCP server	ours	A domain boundary with typed models, full MCP surfaces, shared task and artifact contracts, canonical MCP administration, and only declared additive HTTP projections.
Map server	ours	Earth geography and logistics routing over governed releases; scoped MCP tools, resources, tasks, and Apps own sources, acquisition, activation, rollback, and mobility profiles.
world model	concept	What an agent knows: everything the operation senses, remembers, and decides, fused into one durable, queryable model of reality.
platform store	OSS	SurrealDB — the platform's memory: identity, policy, tasks, artifacts, recordings, map catalogs and releases, agents, audit, and the transactional outbox.
task runtime	ours	The ledger for long work: atomic leases, transitions, results, and declared recovery per operation.
recording forwarder	ours	The producer-local durability edge: loopback Rerun gRPC, bounded persistent queue, OAuth upload, and checkpoint resume.
Recording Hub	ours	The acknowledgement authority: fsynced batch journal, immutable ordered part materialization, and a monotonic checkpoint.
RRD archive shard	format	An immutable footer-indexed shard produced once at an IDR-aligned hour or 192 MiB boundary.
catalog	ours	The card catalog: stable recording identities related to acknowledged parts, archive shards, checkpoints, labels, and lifecycle.
recording server	ours	The reading room: authorized discovery, latest-at / range / dataframe queries, subscriptions, and sealing to an artifact.
SUMO / TraCI	OSS	The traffic simulator and its control socket — a live world producing positions and states to record and reason about.
sumo-mcp	ours	The bridge: the one owner of the simulator's control socket, pushing the world into the record and exposing governed control.
MCP	protocol	The shared protocol for tools, resources, prompts, completions, tasks, subscriptions, notifications, and structured content with declared schemas; one gateway governs the complete surface.
durable task	concept	A tool call you walk away from: the handle returns immediately; the row, lease, and result live in the platform store.
resource · subscribe	concept	A condition that wakes you: subscribing to a resource means the agent is notified the moment it changes.

